# Evaluation runs: **memcached** (memcached-oopsla25)

generated 2026-11-05 from brain-data cfc25f174 2026-11-05 13:40:01 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.0 | 2 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 2 | 3 | 3 | violation | 0.0 | 2 | 1 h, 250 GB | 3 | 9 | brain03, 2026-11-02 |
| 3 | 7 | 7 | violation | 0.0 | 2 | 1 h, 250 GB | 7 | 49 | brain03, 2026-11-02 |

### stateless: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | conforms | 0.0 | 4 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 2 | 3 | 6 | conforms | 0.0 | 2 | 1 h, 250 GB | 3 | 6 | brain03, 2026-11-02 |
| 3 | 7 | 20 | conforms | 0.0 | 2 | 1 h, 250 GB | 7 | 28 | brain03, 2026-11-02 |

### stateless: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | 1 | 1 | brain03, 2026-11-02 |
| 2 | 3 | 3 | conforms | 0.0 | 2 | 1 h, 250 GB | 3 | 6 | brain03, 2026-11-02 |
| 3 | 7 | 7 | conforms | 0.0 | 2 | 1 h, 250 GB | 7 | 28 | brain03, 2026-11-02 |

**Not run (stateless).** the tool ended with an error (exit-101): 4 shared-bucket fifo, 4 lost-set fifo, 4 v3 fifo.

skipped: a smaller size had reached its limit: 5 lost-set fifo, 5 shared-bucket fifo, 5 v3 fifo, 6 lost-set fifo, 6 shared-bucket fifo, 6 v3 fifo, 7 lost-set fifo, 7 shared-bucket fifo, 7 v3 fifo, 8 lost-set fifo, 8 shared-bucket fifo, 8 v3 fifo, 9 lost-set fifo, 9 shared-bucket fifo, 9 v3 fifo.

## stateful

### stateful: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 2 | 3 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 7 | 7 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |

### stateful: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 2 | 3 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 7 | 20 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |

### stateful: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 2 | 3 | 3 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 7 | 7 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-01 |

**Not run (stateful).** the tool ended with an error (exit-101): 4 shared-bucket fifo, 4 lost-set fifo, 4 v3 fifo.

skipped: a smaller size had reached its limit: 5 lost-set fifo, 5 shared-bucket fifo, 5 v3 fifo, 6 lost-set fifo, 6 shared-bucket fifo, 6 v3 fifo, 7 lost-set fifo, 7 shared-bucket fifo, 7 v3 fifo, 8 lost-set fifo, 8 shared-bucket fifo, 8 v3 fifo, 9 lost-set fifo, 9 shared-bucket fifo, 9 v3 fifo.

## offline

### offline: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 3 | 3 | violation | 0.0 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 7 | 7 | violation | 0.0 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |

### offline: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 3 | 6 | conforms | 0.0 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 7 | 20 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |

### offline: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.1 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 3 | 3 | conforms | 0.3 | 4 | 10 min, 16 GB | 0.0 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 7 | 7 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.0 | 0.2 | 0.0 | in-memory | brain05, 2026-11-02 |

**Not run (offline).** skipped: a smaller size had reached its limit: 4 shared-bucket fifo, 4 lost-set fifo, 4 v3 fifo, 5 lost-set fifo, 5 shared-bucket fifo, 5 v3 fifo, 6 lost-set fifo, 6 shared-bucket fifo, 6 v3 fifo, 7 lost-set fifo, 7 shared-bucket fifo, 7 v3 fifo, 8 lost-set fifo, 8 shared-bucket fifo, 8 v3 fifo, 9 lost-set fifo, 9 shared-bucket fifo, 9 v3 fifo.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 171 | 10 min, 16 GB | 0.1 | 0.6 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 3 | violation | 3.1 | 172 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 7 | 7 | violation | 3.4 | 169 | 10 min, 16 GB | 0.0 | 3.4 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | conforms | 2.7 | 158 | 10 min, 16 GB | 0.1 | 2.7 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 6 | conforms | 3.8 | 161 | 10 min, 16 GB | 0.0 | 3.8 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 20 | conforms | 4.3 | 158 | 10 min, 16 GB | 0.0 | 4.3 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 152 | 10 min, 16 GB | 0.1 | 0.6 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 3 | conforms | 3.1 | 166 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 7 | conforms | 4.3 | 174 | 10 min, 16 GB | 0.0 | 4.3 | 0.0 | in-memory | brain01, 2026-11-02 |

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 152 | 10 min, 16 GB | 0.6 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 3 | violation | 2.9 | 154 | 10 min, 16 GB | 2.9 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 7 | violation | 3.1 | 172 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Rust implementation against the Java specification: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | conforms | 0.6 | 152 | 10 min, 16 GB | 0.6 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 6 | conforms | 2.9 | 154 | 10 min, 16 GB | 2.9 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 20 | conforms | 3.1 | 172 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Rust implementation against the Java specification: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 152 | 10 min, 16 GB | 0.6 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 3 | conforms | 2.9 | 154 | 10 min, 16 GB | 2.9 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 7 | conforms | 3.1 | 172 | 10 min, 16 GB | 3.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 3 | 3 | violation | 0.0 | 4 | 1 h, 250 GB | 1 | brain01, 2026-11-02 |
| 3 | 7 | 7 | violation | 0.0 | 4 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |

**cross online, Rust implementation (TraceForge) against the Java specification export: pair shared-bucket, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 3 | 6 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 7 | 20 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

**cross online, Rust implementation (TraceForge) against the Java specification export: pair v3, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 3 | 3 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 7 | 7 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

## twins offline, Rust implementation against its Java twin

**twins offline, Rust implementation against its Java twin: pair lost-set, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 171 | 10 min, 16 GB | 0.6 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 3 | conforms | 3.2 | 172 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 7 | conforms | 3.4 | 169 | 10 min, 16 GB | 3.4 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |

**twins offline, Rust implementation against its Java twin: pair shared-bucket, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | conforms | 2.7 | 158 | 10 min, 16 GB | 2.7 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 6 | 6 | conforms | 3.9 | 161 | 10 min, 16 GB | 3.8 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 20 | 20 | conforms | 4.3 | 158 | 10 min, 16 GB | 4.3 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |

**twins offline, Rust implementation against its Java twin: pair v3, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.7 | 152 | 10 min, 16 GB | 0.6 | 0.3 | 0.1 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 3 | conforms | 3.2 | 166 | 10 min, 16 GB | 3.1 | 0.3 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 7 | conforms | 4.3 | 174 | 10 min, 16 GB | 4.3 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |

## twins offline, Java implementation against its Rust twin

**twins offline, Java implementation against its Rust twin: pair lost-set, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 171 | 10 min, 16 GB | 0.1 | 0.6 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 3 | conforms | 3.1 | 172 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 7 | conforms | 3.4 | 169 | 10 min, 16 GB | 0.0 | 3.4 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Java implementation against its Rust twin: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | conforms | 2.7 | 158 | 10 min, 16 GB | 0.0 | 2.7 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 6 | 6 | conforms | 3.8 | 161 | 10 min, 16 GB | 0.0 | 3.8 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 20 | 20 | conforms | 4.3 | 158 | 10 min, 16 GB | 0.1 | 4.3 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Java implementation against its Rust twin: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 152 | 10 min, 16 GB | 0.3 | 0.6 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 3 | 3 | conforms | 3.1 | 166 | 10 min, 16 GB | 0.3 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 7 | 7 | conforms | 4.3 | 174 | 10 min, 16 GB | 0.2 | 4.3 | 0.0 | in-memory | brain01, 2026-11-02 |

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 156 | 10 min, 16 GB | 0.6 | 0.1 | 0.0 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | conforms | 2.7 | 164 | 10 min, 16 GB | 2.7 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.7 | 148 | 10 min, 16 GB | 0.6 | 0.2 | 0.1 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** refused (an input missing at run time): 2 fifo lost-set, 2 fifo shared-bucket, 2 fifo v3, 3 fifo shared-bucket, 3 fifo lost-set, 3 fifo v3.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 156 | 10 min, 16 GB | 0.1 | 0.6 | 0.0 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | conforms | 2.7 | 164 | 10 min, 16 GB | 0.2 | 2.7 | 0.0 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.7 | 148 | 10 min, 16 GB | 0.2 | 0.6 | 0.1 | in-memory | brain02, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: 2 fifo lost-set, 2 fifo shared-bucket, 2 fifo v3, 3 fifo shared-bucket, 3 fifo lost-set, 3 fifo v3.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.7 | 149 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 3 | | | | | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 7 | | | | | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | conforms | 3.3 | 152 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 6 | | | | | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 20 | | | | | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.6 | 152 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 3 | | | | | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 7 | | | | | 1 h, 250 GB | | brain02, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **lost-set**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 2.1 | 2 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **shared-bucket**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **v3**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** skipped: a smaller size had reached its limit: 2 fifo lost-set, 2 fifo shared-bucket, 2 fifo v3, 3 fifo shared-bucket, 3 fifo lost-set, 3 fifo v3.